

Documentation: User Authentication and Product Display Workflow

Overview

This document outlines the process and workflow for implementing user authentication using Firebase and displaying product details upon successful login. The system includes two main forms: **Login** and **Registration**, each with form validation to ensure data integrity.

Components

1. Firebase Integration

- **Purpose:** Handle user authentication (login and registration).
- **Actions:**
 - Connect the application to Firebase services.
 - Use Firebase Authentication to manage user accounts.

2. Registration Process

- **Trigger:** User selects the registration option.
- **Steps:**
 1. **Form Validation:** Ensure all fields (e.g., email, password) are filled.
 - Display error messages for empty fields.
 2. **Firestore Registration:**
 - If validation passes, send the registration data to Firestore.
 - Handle success or error responses (e.g., account already exists).

3. Login Process

- **Trigger:** User selects the login option.
- **Steps:**
 1. **Form Validation:** Ensure all fields (e.g., email, password) are filled.
 - Display error messages for empty fields.
 2. **Firestore Authentication:**
 - If validation passes, authenticate user credentials with Firestore.
 - Handle success or error responses (e.g., incorrect password).

4. Post-Login Product Display

- **Trigger:** Successful login.
- **Steps:**
 1. Dynamically generate product details using the Document Object Model (DOM).
 2. Display products on the user interface.

Error Handling

- **Empty Fields:** Prompt user to complete all required fields.
- **Registration Errors:** Notify if the email is already registered.
- **Login Errors:** Alert on incorrect credentials.

User Flow

1. **New User:**
 - Opens the app → Selects **Register** → Completes form → Submits → On success, proceeds to product display.
2. **Returning User:**
 - Opens the app → Selects **Login** → Completes form → Submits → On success, proceeds to product display.

Security Considerations

- Use secure password handling practices.
- Validate user inputs to prevent injection attacks.
- Implement Firebase security rules for data access.

Future Enhancements

- Add password reset functionality.
- Implement email verification.
- Enhance product display with sorting and filtering options.